

Geophysical Methods→ Different ways to explore the subsurface

Each method investigates a different physical property of subsurface materials:

- ERT → Electrical Resistivity:
 - indicates the material's ability to conduct electric current
 - typically used for groundwater studies, contaminant mapping, and geological structures
 - penetration depth: up to several tens of meters
- EM (Induction) → Electrical Conductivity:
 - measures response to time-varying electromagnetic fields
 - used for mapping conductivity contrasts, mineral exploration, salinity studies
 - penetration: depends on frequency, from a few meters to hundreds
- GPR → Dielectric Permittivity:
 - reflects the ability of materials to store energy in an electric field
 - used for shallow investigations: utility detection, archaeology, stratigraphy
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Until now, we have focused on how materials interact with electric or electromagnetic fields.

But what haven't we explored yet?

The magnetic response of materials.

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The Earth has a natural magnetic field — the geomagnetic field

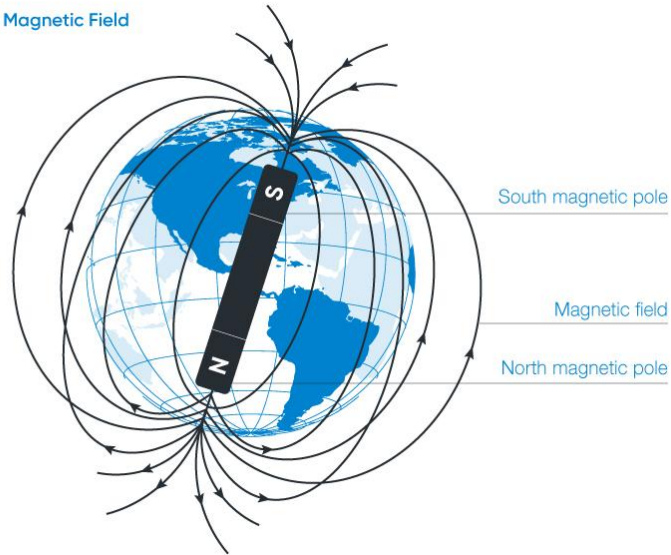
Magnetic fundamentals

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Earth's Magnetic Field



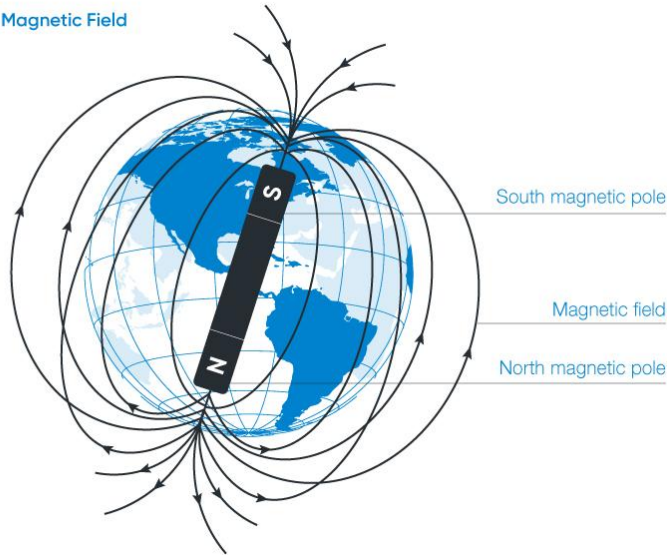
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Earth's Magnetic Field



This field is generated by electric currents flowing in the liquid outer core, which is composed mostly of molten iron and nickel.

According to Ampère's Law, a moving electric charge (or current) generates a magnetic field.

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

At the global scale, we often approximate the Earth's magnetic field as if it were generated by a giant bar magnet, or magnetic dipole.

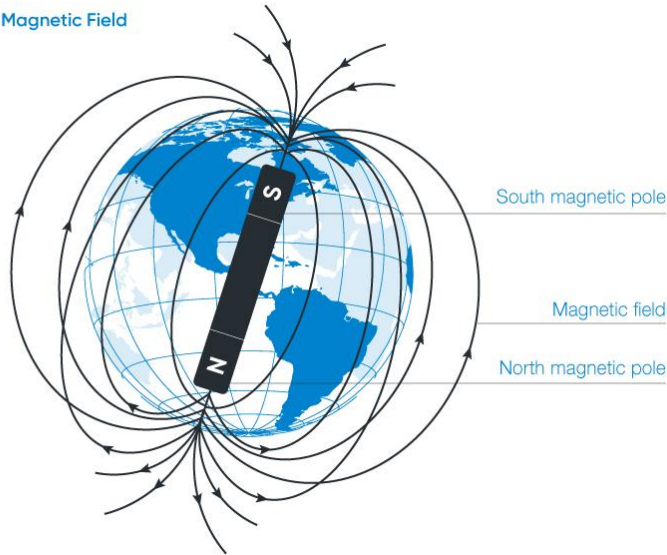
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This field can be disturbed by subsurface bodies with magnetic properties.

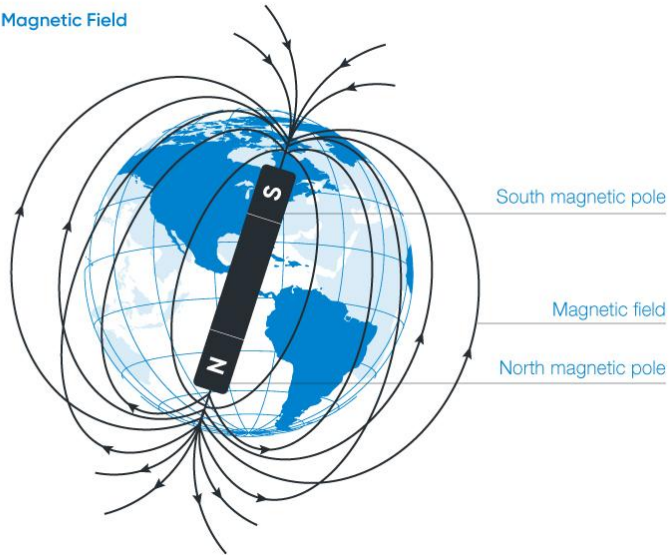
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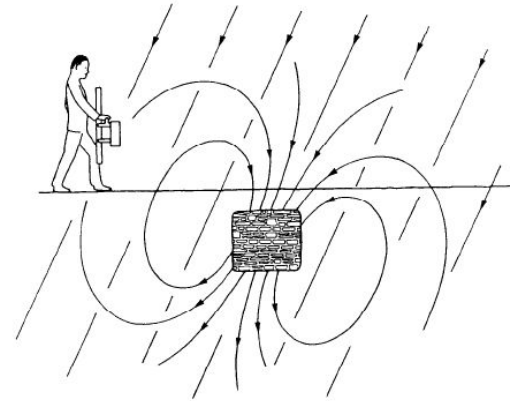
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Earth's Magnetic Field



We can detect these disturbances and infer subsurface structure.



Magnetic anomalies depend on the properties of the source body:

- Anomalies may be caused by:
 - Mineral deposits (e.g., magnetite-rich rocks)
 - Buried ferrous objects (metal debris, archaeol. artifacts)
- Higher magnetic **susceptibility** leads to stronger anomalies

Magnetic Susceptibility of Materials

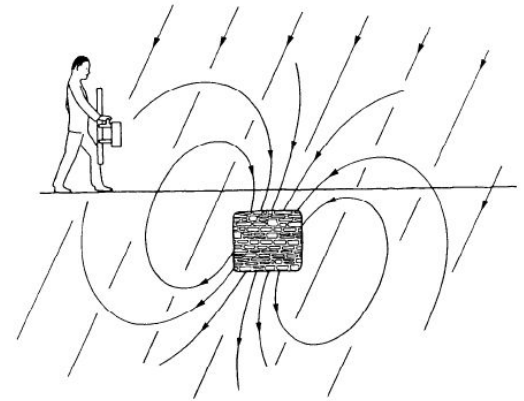
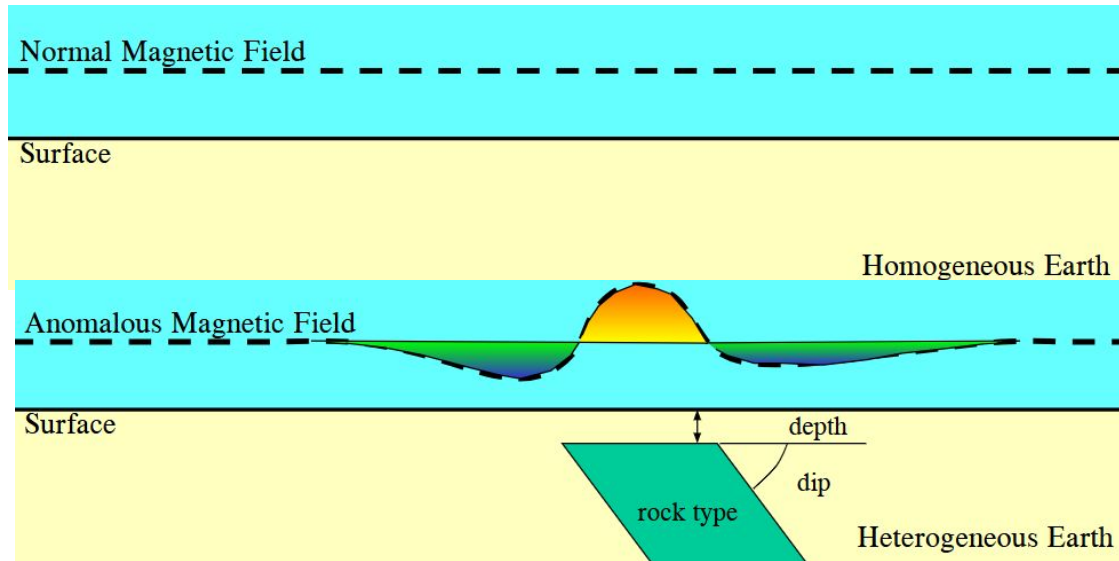
- **Material Susceptibility x 10^3 (SI)***

- Air ~0
- Quartz -0.01
- Rock Salt -0.01
- Calcite -0.001 - 0.01
- Sphalerite 0.4
- Pyrite 0.05 – 5
- Hematite 0.5 – 35
- Ilmenite 300 – 3500
- Magnetite 1200 - 19,200
- Limestones 0 – 3

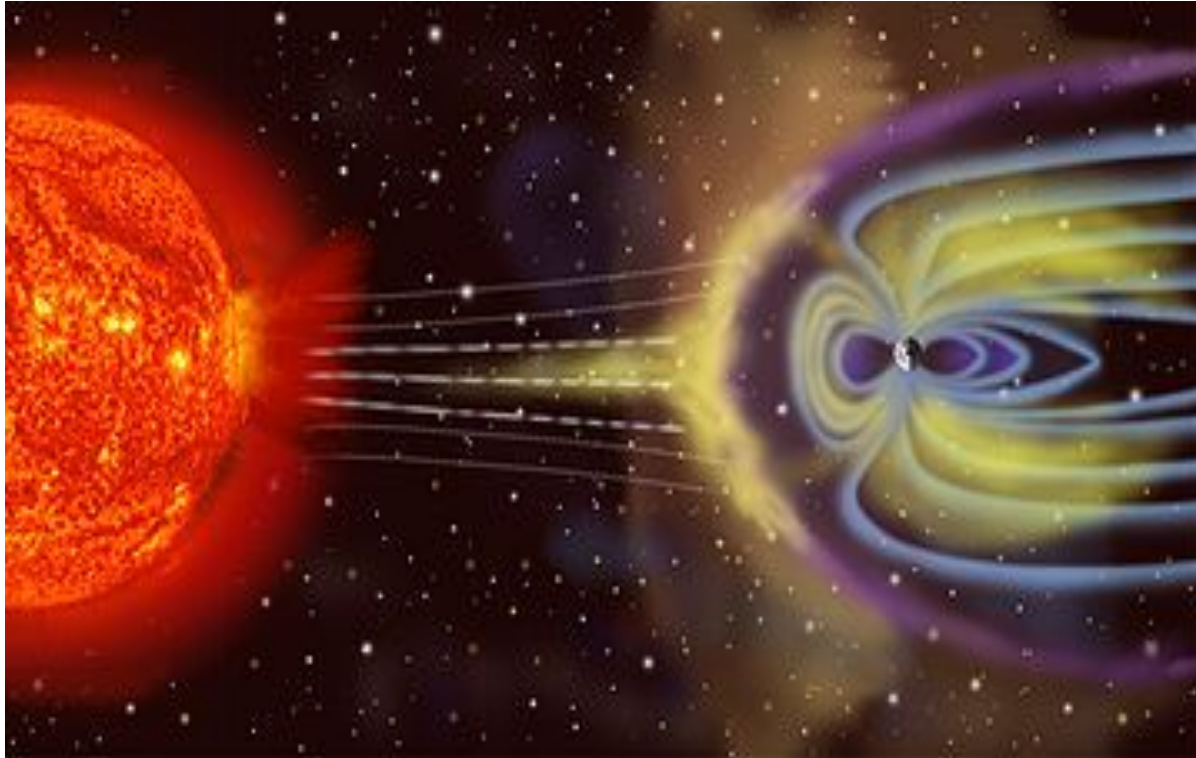
- **Material Susceptibility x 10^3 (SI)***

- Sandstones 0 – 20
- Shales 0.01 – 15
- Schist 0.3 – 3
- Gneiss 0.1 – 25
- Slate 0 – 35
- Granite 0 – 50
- Gabbro 1 – 90
- Basalt 0.2 – 175
- Peridotite 90 - 200

- In a small area the external magnetic field over a homogeneous earth is constant.
- The external field is distorted when there is a susceptibility contrast in the subsurface
- From the shape of the distortion (anomaly) we can deduce some attributes of the source body.



Diurnal variations



These are variations in the magnetic field that occur over the course of a day and are related to variations in the Earth's external magnetic field.

Diurnal variations in the Earth's magnetic field result from the interaction of ionospheric electric currents, the geomagnetic field, and the solar wind.

This variation are normally in the order of 20 to 50 nT per day but can reach up to 200 nT.

The diurnal variations should be accounted for when conducting exploration magnetic surveys.

On the surface of the earth the magnetic field can be broken into three separate components

Main Field

- Largest and most stable component.
- Generated by electrical currents in the Earth's outer core.
- Modeled by the IGRF. the International Geomagnetic Reference Field.
- Ranges from ~25,000 to 65,000 nT.
- Changes slowly (timescale of years).

External Magnetic Field

- Generated by sources external to the Earth.
- Changes quickly (timescale of minutes).
- Typically ranges from 10 to 50 nT.

Local Magnetic Field (Anomalous Field)

- Produced by near-surface sources like rocks, buried objects, minerals, or archaeological remains.
- Main target in geophysical prospecting.
- Typically ranges from ~10 to 10,000 nT.

Ground Magnetometer System

Highly sensitive scalar sensor used to measure the **total magnetic field intensity**

$$B_{\text{measured}} = |\vec{B}| = \sqrt{B_x^2 + B_y^2 + B_z^2} \quad B_{\text{measured}} = B_{\text{Earth}} + B_{\text{anomaly}} + B_{\text{diurnal}} + \varepsilon$$

- High-resolution measurement of the **geomagnetic field** (sensitivity: ~0.02 nT).
- Use as a **base station** to monitor diurnal variations (if stationary).
- Detection and mapping of **large-scale geological structures**, metalliferous mineral lodes, and archaeological remains.
- Locating **metallic objects**, buried pipelines, **military ordnance**, and toxic waste drums.



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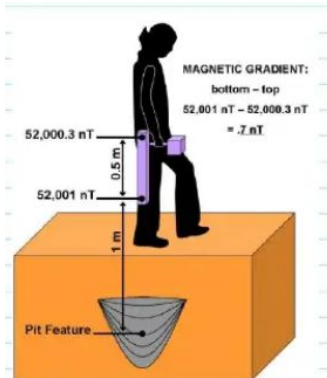
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Vertical Magnetic Gradient →

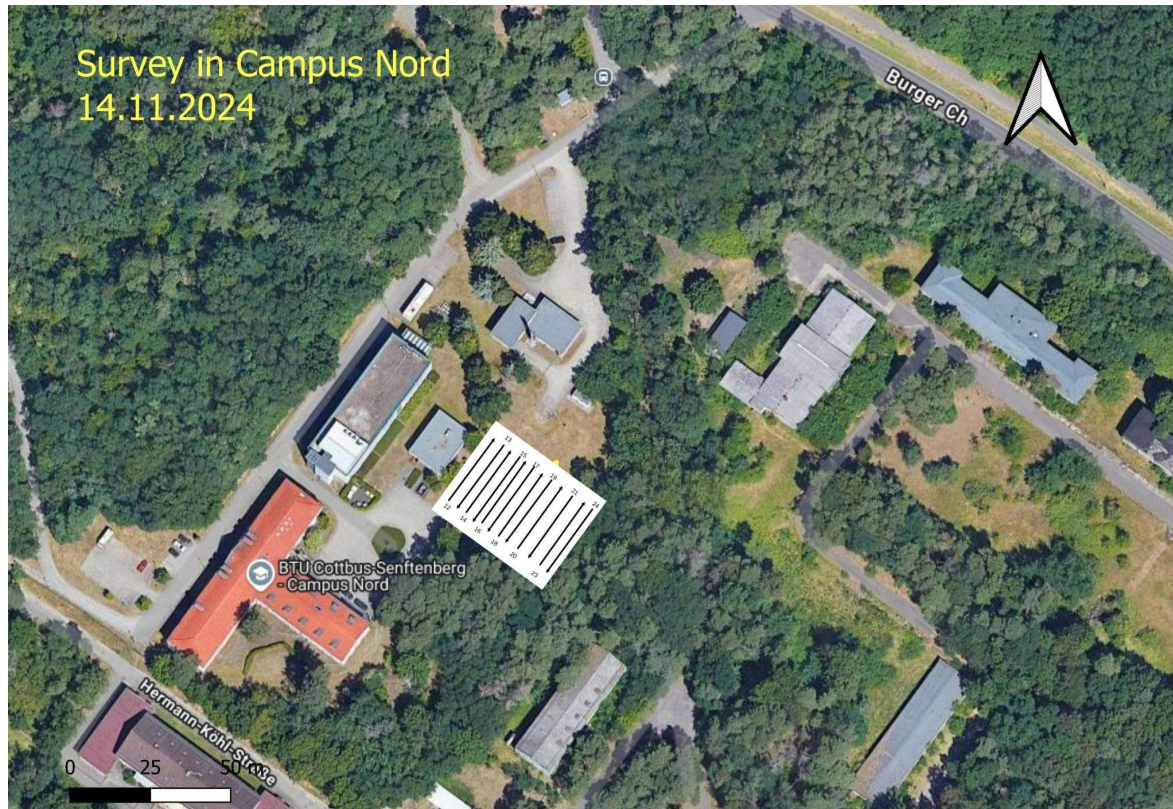
$$\Delta B_z = \frac{B_{\text{top}} - B_{\text{bottom}}}{\Delta z}$$

- B_{top} : field measured by the upper sensor,
- B_{bottom} : field measured by the lower sensor,
- Δz : vertical distance between both sensors.

- Enhances the detection of shallow sources.
- improves the resolution of the anomaly shape and boundaries

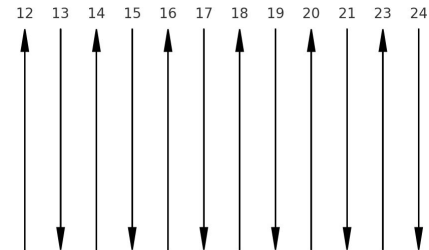


Datos bien medidos, sondeo viejo. se midio

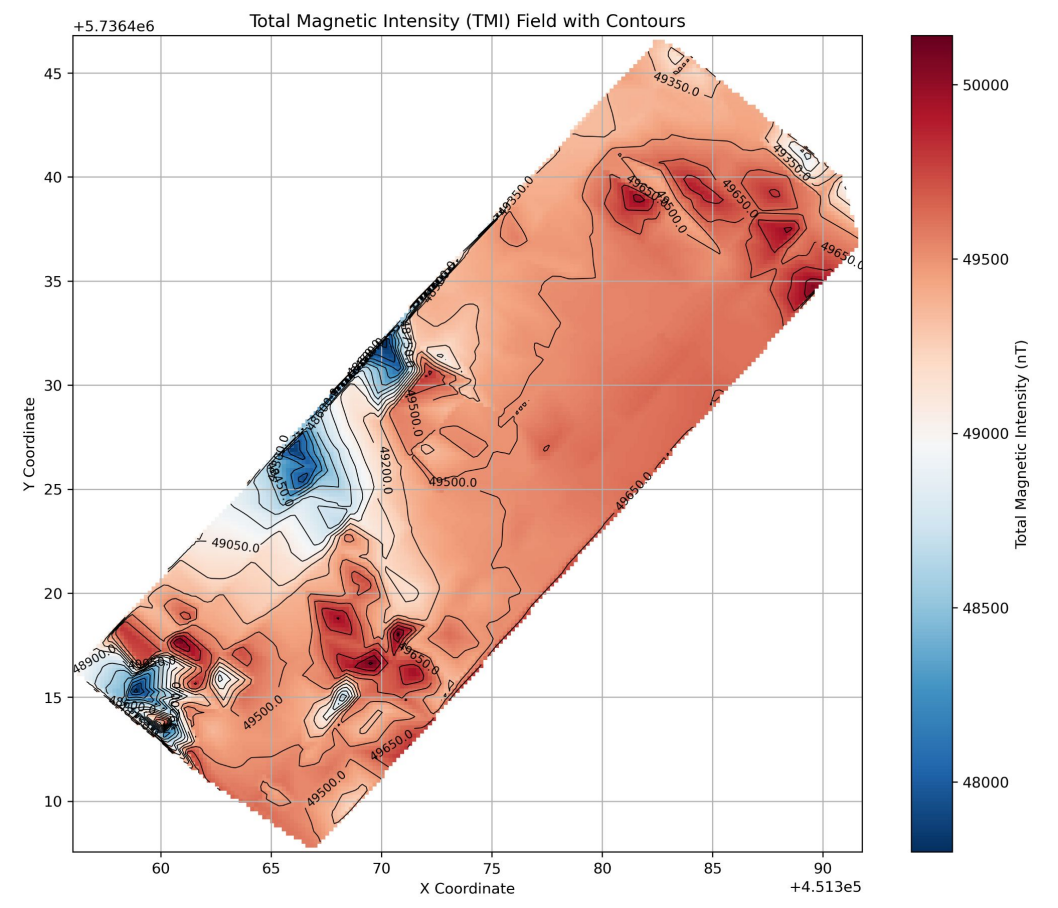
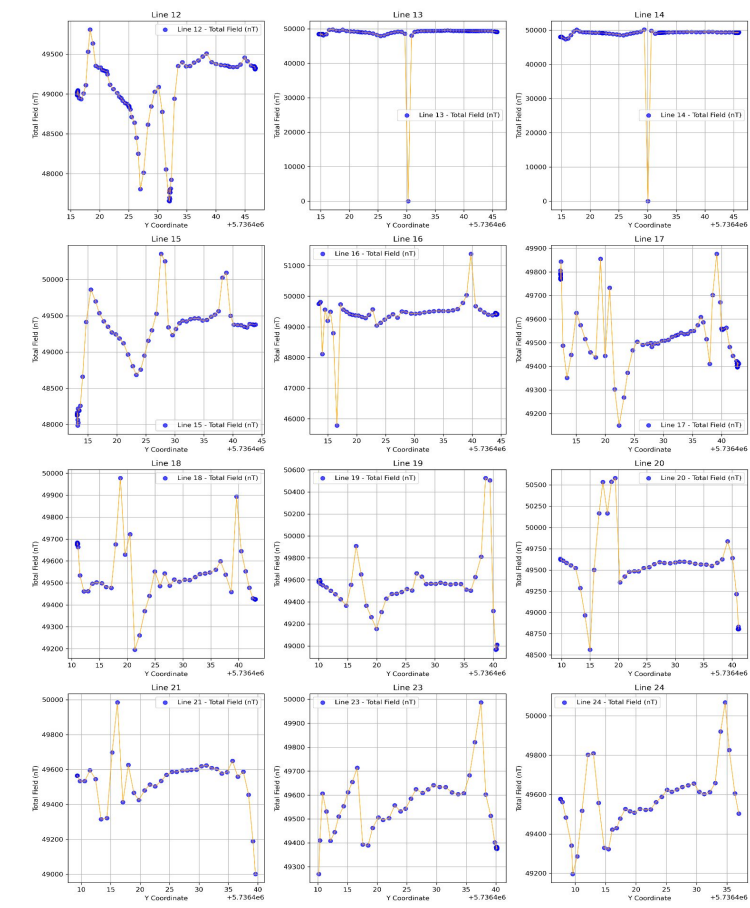


1 - campo magnético
total
2 - gradiente campo
magnético

survey lines



Total magnetic field results



Gradient magnetic field results

